AI System for Detecting Silent Failures in ML Pipelines: Implementation and Experimental Analysis

Dishan Hari Seelam, Harsha M Gowda, Thejas Naika C, and Revathy B D
Department of CSE (AI&ML), Vidyavardhaka College of Engineering, Mysuru, India

Abstract - This report implements and evaluates an AI-based silent failure detection system for an e-commerce fraud-classification pipeline. Building on the surveyed need for integrated ML observability, the system combines feature validation, Population Stability Index (PSI) and Kolmogorov-Smirnov (KS) drift detection, Isolation Forest anomaly detection, training-serving skew checks, and a unified health score. A 2,000-record synthetic e-commerce dataset with a 2.95% fraud rate was used to train a Random Forest classifier. Threshold selection improved fraud recall from 16.67% to 41.67% on the held-out test set, while the controlled drift experiment detected four shifted features with large PSI values. The implementation demonstrates how multiple weak monitoring signals can be converted into an actionable pipeline-level assessment without requiring immediate production labels.

Index Terms - ML observability, silent failure, data drift, PSI, KS test, Isolation Forest, training-serving skew, MLOps.

I. INTRODUCTION

Production machine learning systems can continue to return predictions after their assumptions have failed. The submitted survey identifies data drift, feature corruption, anomalous prediction behavior, and training-serving skew as silent failures because they are not necessarily accompanied by runtime exceptions. Individual monitoring methods exist, but the survey concludes that they are often fragmented and offer limited diagnosis [1]-[10].

This project implements the survey's proposed monitoring direction in an e-commerce fraud setting. Its contribution is not a new base classifier; rather, it is a unified decision layer that consolidates heterogeneous monitoring signals into a health score, severity, failure type, and recommended action.

II. RELATED WORK AND DESIGN RATIONALE

Reference	Relevant contribution	Project response
Shankar et al. [4]	Detection, diagnosis, and reaction; PSI and KS for unlabeled monitoring.	Implements PSI/KS batch drift tests and presents feature-level results.
Leest et al. [3]	Industry monitoring remains fragmented across metrics and metadata.	Unifies validation, drift, anomaly, and skew findings in one dashboard score.
Dara and Cavus [8]	Explainable drift diagnosis improves retraining decisions.	Stores feature-level drift results; SHAP is retained as the next root-cause extension.
Reddy [9]; Sana [10]	Anomaly monitoring requires historical context and can create false positives.	Uses Isolation Forest as an additional signal rather than a sole failure decision.

Table I. *Alignment of the implementation with the studies reviewed in the submitted survey.*

The framework deliberately prioritizes label-independent signals. This addresses the survey's observation that production ground truth may be unavailable or delayed. However, the current evaluation is synthetic and should be treated as implementation validation rather than proof of performance in a live business environment.

III. SYSTEM ARCHITECTURE

The deployed FastAPI service validates each incoming transaction, derives the same timestamp features used during training, produces a fraud probability, applies the tuned alert threshold, and evaluates an Isolation Forest anomaly detector. Batch monitoring compares current features with the training reference distribution using PSI and KS. A training contract checks feature order and preprocessing version. The dashboard exposes all saved artifacts and a unified health engine.

The health score starts at 100 and deducts 30 points for failed validation, up to 30 for drift, 20 for a prediction anomaly, and 20 for training-serving skew. Scores of 85 or more are healthy; 60-84 require attention; below 60 requires intervention.

IV. EXPERIMENTAL SETUP

The experiment uses 2,000 synthetic e-commerce transactions. The target is *is_fraudulent*; 2.95% of transactions are positive. The data split contains 1,200 training, 400 validation, and 400 test records. Categorical fields are one-hot encoded and numerical fields use median imputation within one scikit-learn pipeline, preventing a preprocessing mismatch between model training and normal API inference.

A Random Forest classifier is trained with class weighting because fraud is rare. The decision threshold is selected on the validation partition using F2 score, which weights recall twice as strongly as precision. The test partition is never used for selecting the threshold.

A controlled production batch of 200 samples is created from the reference data. The experiment multiplies order value by 3.5, multiplies unit price by 2.8, increases customer age by 25 years (capped at 100), and adds 900 km to shipping distance. This known intervention enables deterministic drift verification.

V. MODEL RESULTS

Threshold tuning: test-set metrics

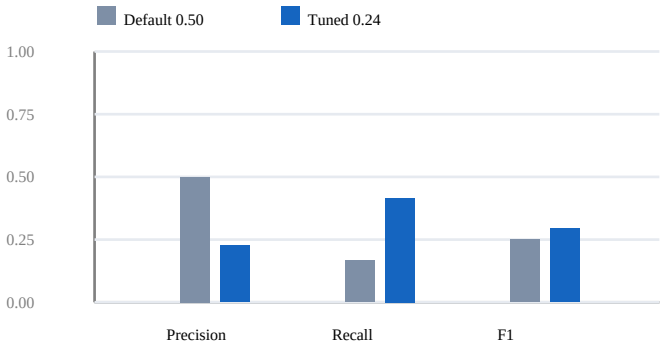

Fig. 1. Test-set precision, recall, and F1 before and after validation-selected threshold tuning.

T	P	R	F1	CM (TN, FP, FN, TP)
0.50	0.500	0.167	0.250	386, 2, 10, 2
0.24	0.227	0.417	0.294	371, 17, 7, 5

Table II. *Test-set threshold T, precision P, recall R, F1, and confusion-matrix counts. TN, FP, FN, and TP denote true negative, false positive, false negative, and true positive.*

The selected threshold is 0.24. Fraud recall rises from 16.67% to 41.67%, detecting 5 of 12 test fraud cases rather than 2 of 12. This benefit comes with lower precision (22.73%) and 17 false positives. In a fraud-review workflow, this is an explicit operating trade-off: more suspected orders are routed to review in return for fewer missed fraud cases. The threshold-independent ROC-AUC is 0.705 and average precision is 0.300; these moderate values are appropriate to report alongside accuracy because the class distribution is highly imbalanced.

VI. DRIFT-DETECTION EXPERIMENT

Injected production drift: PSI by affected feature

Fig. 2. PSI values for intentionally shifted features in the 200-record production batch.

The drift detector returned **drift_detected** for the 200-record batch. The affected features were customer age, unit price, order value, shipping distance km. The largest PSI values were shipping distance (11.0346) and customer age (10.5733); both exceed the configured 0.20 warning threshold by a substantial margin. Their KS p-values are effectively zero after rounding, consistent with a changed distribution. Features not deliberately modified remained below the PSI threshold and had non-significant KS results, supporting the detector's specificity in this controlled experiment.

VII. ANALYSIS OF MONITORING COVERAGE

Failure signal	Implementation evidence
Feature validation	Rejects unseen categorical values and implausible numeric ranges before inference.
Data drift	PSI and KS identify shifted feature distributions in a batch.
Prediction anomaly	Isolation Forest evaluates probability and selected transaction features.
Training-serving skew	Checks feature list, order, and preprocessing version against training contract.
Unified health	Combines all signals into 0-100 score and severity.

Table III. *Implemented monitoring coverage and the associated remediation path.*

This design directly addresses the survey's lack-of-unification gap. A drift alert alone can be noisy; an anomaly alone cannot prove data shift; and a skew check cannot identify a corrupt input. The health engine preserves each module's evidence but provides one operational status for a reviewer. The weights are transparent and can be adjusted based on business risk or alert burden.

VIII. LIMITATIONS AND FUTURE WORK

The results should not be overgeneralized. The e-commerce dataset is synthetic, only 59 of 2,000 observations are fraudulent, and the test set contains 12 fraud cases. Confidence intervals and external benchmark comparisons are therefore not claimed. PSI/KS detection measures distribution change, not necessarily concept drift or model performance degradation. The Isolation Forest is calibrated on synthetic baseline behavior. In production, monitored batches, feedback labels, threshold review, alert history, and periodic retraining policies are required.

The survey proposes SHAP-based root-cause analysis. The current project preserves this as the next extension: when health becomes warning or critical, SHAP can explain the classifier's top contributing features for the flagged transaction or batch. Additional future work includes automated alerting, a database-backed event log, time-series health trends, and evaluation on a real labeled e-commerce fraud dataset.

IX. CONCLUSION

The implemented system transforms the survey's unified-monitoring proposal into a functioning MLOps prototype. It demonstrates validation, anomaly, drift, and skew monitoring in one application and produces an interpretable health assessment. Threshold tuning improved fraud recall from 16.67% to 41.67%, and the controlled experiment detected all four intentionally shifted features. The main value is therefore not a claim of a new classifier, but a transparent integrated workflow for identifying and responding to silent failures before they become undetected operational risk.

REFERENCES

[1] R. Pathak et al., "Detecting Silent Failures in Multi-Agentic AI Trajectories," arXiv preprint arXiv:2505.14144, 2025.

[2] Y. Yang, "Taming Silent Failures: A Framework for Verifiable AI Reliability," arXiv preprint, 2025.

[3] J. Leest, "Monitoring and Observability of Machine Learning Systems: Current Practices and Gaps," arXiv preprint arXiv:2505.19395, 2025.

[4] R. Shankar, R. Arora, A. Jain, and A. Sinha, "Towards Observability for Production Machine Learning Pipelines," in Proc. IEEE Int. Conf. Big Data, 2022.

[5] Z. Jiang et al., "Training with Confidence: Catching Silent Errors in Deep Learning Training with Automated Proactive Checks," Proc. ACM Program. Lang., 2025.

[6] H. Zhao, Y. Guo, and X. Li, "Lightweight Detection of Silent Data Corruption in Distributed Deep Learning," in Proc. IEEE IPDPS, 2025.

[7] K. Nandamuri, "A Comprehensive Guide to Monitoring and Observability in Machine Learning Infrastructure: From Metrics to Implementation," arXiv preprint, 2025.

[8] E. Dara and N. Cavus, "From XAI to MLOps: Explainable Concept Drift Detection with Profile Drift Detection," Expert Systems with Applications, vol. 271, 2025.

[9] V. Reddy, "Machine Learning-Driven Anomaly Detection in CI/CD Pipelines for Financial Applications," International Journal of Computer Applications, vol. 187, no. 12, pp. 15-24, 2024.

[10] S. Sana, "Developing an AI-Driven Anomaly Detection System for Cloud Data Pipelines: Minimizing Data Quality Issues by 40%," International Journal of Advanced Computer Science and Applications, vol. 16, no. 3, pp. 221-231, 2025.